

IB MATH AA SL

Lesson 9

Proof & Deductive Reasoning

Number & Algebra · 60-minute lesson

Syllabus SL 1.6 · Property of Lynda Badmus Education



Do Now — recall

ON YOUR WHITEBOARD

1. Write an even number in terms of n .
2. Write an odd number in terms of n .
3. Is the sum of two even numbers always even?

Answers (click to reveal)

1. $2n$
2. $2n + 1$
3. Yes — $2a + 2b = 2(a+b)$

What we're learning today

LEARNING OBJECTIVES

- 1 Understand**
what a mathematical proof requires.
- 2 Use**
algebra to represent general numbers.
- 3 Construct**
a direct (deductive) proof.
- 4 Recognise**
proof by counterexample.

SUCCESS CRITERIA — I CAN...

- ✓
write even/odd/consecutive numbers algebraically
- ✓
argue generally, not from examples
- ✓
set out a clear step-by-step proof
- ✓
disprove a statement with one counterexample

Representing numbers

ALGEBRAIC FORMS

$$\text{even} = 2n, \quad \text{odd} = 2n + 1$$

Consecutive

$n, n+1, n+2, \dots$

Multiple of 3

$3n$

Key idea

a letter stands for ANY integer — that makes it general.

WHAT COUNTS

Examples are NOT a proof.

A general algebraic argument IS.

Conclude

end by stating the result is proven.

A direct proof

QUESTION

Prove that the sum of two consecutive integers is always odd.

SOLUTION

Step 1 let the integers be n and $n+1$:

Step 2 add them:

$$n + (n+1) = 2n+1$$

Step 3 $2n+1$ is odd for any integer n .

Therefore the sum of two consecutive integers is always odd. ■

Your turn — prove it

ANSWER IN YOUR BOOK

1. Prove the sum of two even numbers is even.
2. Prove the sum of three consecutive integers is a multiple of 3.

Answers (click to reveal)

1. $2a+2b = 2(a+b) \rightarrow \text{even}$
2. $n+(n+1)+(n+2)=3n+3=3(n+1) \rightarrow \text{multiple of 3}$

Counterexamples

DISPROVING A CLAIM

one counterexample disproves a statement

To disprove 'all ...', find ONE case that fails.

You don't need many — a single one is enough.

EXAMPLE

Claim

' n^2 is always even.'

Test $n = 3$

$3^2 = 9$, which is odd.

So

the claim is false — disproved.

Use a counterexample

QUESTION

Disprove: “Every prime number is odd.”

SOLUTION

Step 1 test small primes:

Step 2 2 is prime, but 2 is even.

Step 3 so not every prime is odd.

The single example $n = 2$ disproves the statement. ■

Exam-style question

- (a) Prove that the product of two odd numbers is odd. [4]
(b) Disprove: “ $n^2 - n + 11$ is prime for all integers n .” [2]

MARK SCHEME — reveal after students attempt (tutor only)

- (a) Odd $\times$ odd: $(2a+1)(2b+1) = 4ab+2a+2b+1$ (M1)(A1) $= 2(2ab+a+b)+1$ (A1) which is odd (R1)
(b) $n = 11$: $11^2 - 11 + 11 = 121 = 11 \times 11$, not prime (M1)(A1)

Common mistakes



Examples $\neq$ proof

Showing it works for 3 numbers does not prove it for all.



Define your letters

State 'let n be an integer' so the argument is general.



Odd form

An odd number is $2n+1$ (or $2n-1$), never just ' n '.



Conclude

Finish with a clear concluding statement.

Mixed practice

QUESTIONS

1. Prove $2n + 2n$ is even.
2. Prove the sum of two odd numbers is even.
3. Disprove: 'all multiples of 3 are odd.'
4. Write three consecutive even numbers.

Answers (click to reveal)

1. $4n = 2(2n) \rightarrow$ even
2. $(2a+1)+(2b+1)=2(a+b+1) \rightarrow$ even
3. 6 is a multiple of 3 but even
4. $2n, 2n+2, 2n+4$

Plenary & exit ticket

KEY FORMULAS

$$even = 2n$$

$$odd = 2n + 1$$

EXIT TICKET — before you leave

1. Write an odd number algebraically.
2. How many counterexamples disprove a claim?
3. Is 2 a counterexample to 'all primes are odd'?

Answers (click to reveal)

1. $2n + 1$
2. Just one
3. Yes

Self-assess:
• need more help
• nearly there
• confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Proof & reasoning.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 10

End of Topic Test — Number & Algebra. Bring everything from Lessons 2–9.